

Assignment 2

Q1. Demonstrate the following network topology through simulation [Star, Ring, Bus, Mesh, Hybrid]

1. Star Topology

- Create a central **Cisco 2960 switch**.
- Connect multiple PCs to the switch using **Copper Straight-Through cables**.
- Configure the IP address and subnet mask for each PC.
- Test communication between the connected PCs using **Simple PDU** in Simulation Mode.
- Observe the packet transmission through the central switch.

2. Ring Topology

- Create multiple **Cisco 2960 switches** and connect them in a closed-loop arrangement.
- Connect PCs to the respective switches.
- Use **Copper Straight-Through cables** for PC-to-switch connections.
- Configure IP addresses and subnet masks for all PCs.
- Test packet transmission between different PCs using **Simple PDU**.

3. Bus Topology

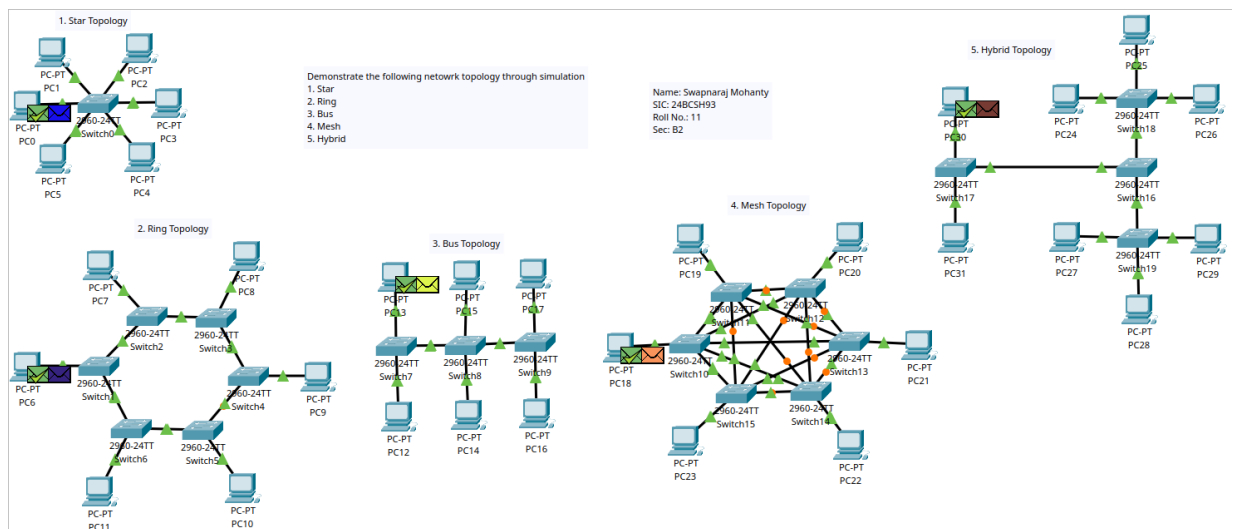
- Arrange multiple switches along a common communication path.
- Connect PCs to the respective switches.
- Configure the IP address and subnet mask for each PC.
- Use **Simple PDU** to test communication between PCs.
- Observe the transmission of packets through the common network path.

4. Mesh Topology

- Create multiple **Cisco 2960 switches**.
- Connect the switches with multiple links so that there are several possible paths between the switches.
- Connect PCs to the switches.
- Configure IP addresses and subnet masks for all PCs.
- Use **Simulation Mode** and **Simple PDU** to observe packet transmission through the different available paths.

5. Hybrid Topology

- Combine two or more different network topologies in a single network.
- In the shown topology, multiple switches are interconnected to form a combined network structure.
- Connect PCs to the respective switches.
- Configure IP addresses and subnet masks for all PCs.
- Test communication between PCs using **Simple PDU**.
- Observe packet transmission across the combined topology.



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